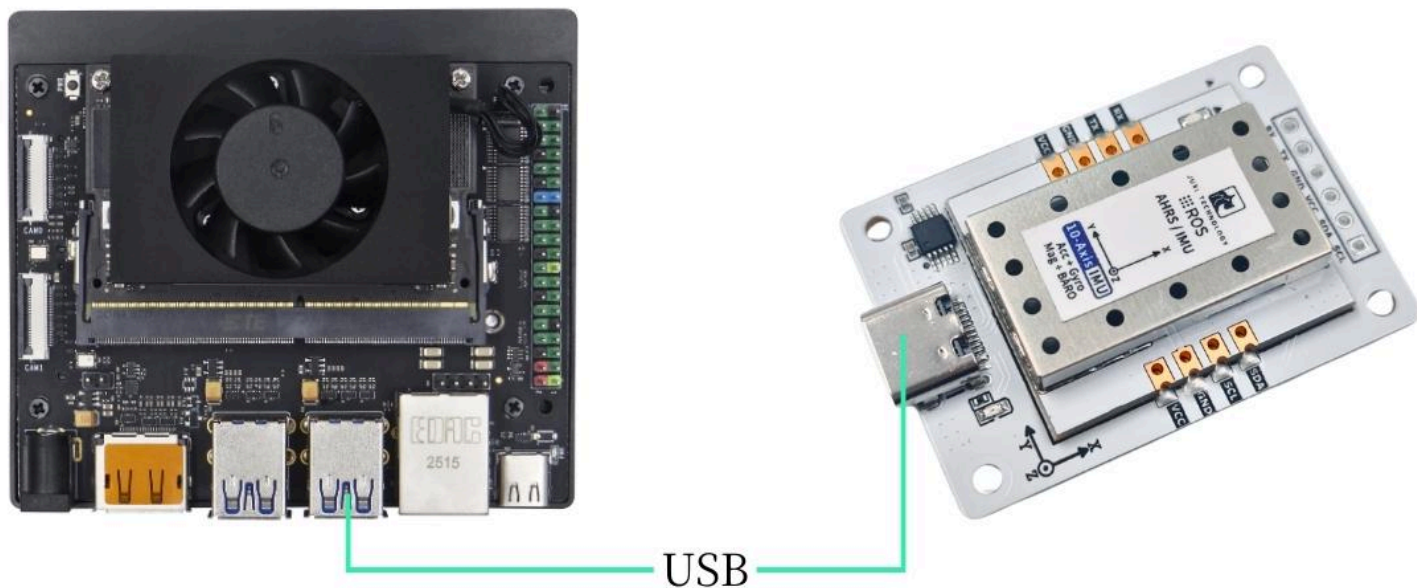


Jetson Series

1. Connect the device

This tutorial takes the Jetson Orin NX motherboard as an example.

Connect the IMU attitude sensor to the USB port of the main controller via a Type-C cable.



2. Check device status

View device ID

代码块

```
1  lsusb
```

View Device Number

代码块

```
1  ls -l /dev/ttyU*
```

```
juxi@juxi-desktop: ~  
(base) juxi@juxi-desktop:~$ lsusb  
Bus 002 Device 002: ID 2109:0822 VIA Labs, Inc. USB3.1 Hub  
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub  
Bus 001 Device 003: ID 0bda:c822 Realtek Semiconductor Corp. Bluetooth Radio  
Bus 001 Device 004: ID 093a:2510 Pixart Imaging, Inc. Optical Mouse  
Bus 001 Device 005: ID 1a86:7523 QinHeng Electronics CH340 serial converter  
Bus 001 Device 006: ID 0c45:7638 Microdia Akko Keyboard  
Bus 001 Device 002: ID 2109:2822 VIA Labs, Inc. USB2.0 Hub  
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub  
(base) juxi@juxi-desktop:~$ ls -l /dev/ttyU*  
crw-rw---- 1 root dialout 188, 0 Feb  5 10:15 /dev/ttyUSB0  
(base) juxi@juxi-desktop:~$
```

Set Port Mapping

代码块

```
1  # 防止插拔后端口变更, 请设置端口映射  
2  sudo gedit /etc/udev/rules.d/99-serial-imu.rules  
3  
4  # 如出现没有gedit命令相关内容, 先下载安装  
5  sudo apt install gedit  
6  
7  # 填写映射内容  
8  KERNEL=="ttyUSB*", ATTRS{idVendor}=="1a86", ATTRS{idProduct}=="7523",  
   MODE=="0777", SYMLINK+="imu-serial"  
9  
10 # 参数说明:  
11 `--mode`: 通信模式, 可选值为`serial`(串口)或`i2c`  
12 `--port`: 串口名(如`/dev/ttyUSB0`)或I2C端口号(如`7`)  
13 `--rate`: 数据打印频率(Hz), 默认10Hz  
14 `--debug`: 启用调试模式, 显示详细信息  
15  
16 # 保存退出, 运行命令使规则生效  
17 sudo udevadm trigger  
18 sudo service udev reload  
19 sudo service udev restart  
20  
21 # 验证  
22 ll /dev/imu-serial  
23
```

```
24 # 输出示例:
25 lrwxrwxrwx 1 root root 7 1月 22 10:00 /dev/imu-serial -> ttyUSB0
```

3. Install the driver library

3.1 Install the Python libraries required for the code

代码块

```
1 sudo apt update
2 sudo apt install -y python3-serial
3 sudo apt install -y python3-smbus2
```

3.2 Transfer Files

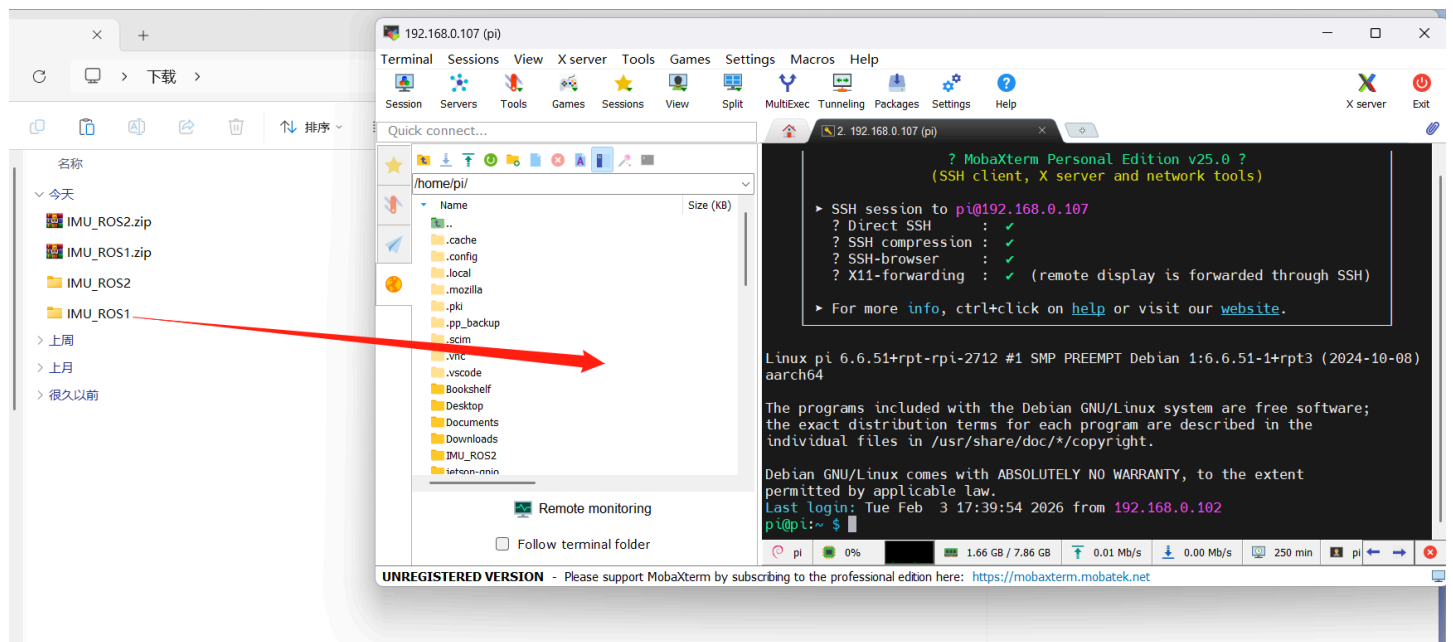


IMU_ROS2.zip
357.96KB



Friends who are not yet familiar with using MobaXterm to transfer files, please refer to the following webpage for detailed installation and operation methods of MobaXterm: [🌐 文件远程传输](#)

Drag the extracted files onto Jetson via MobaXterm software.



4. View IMU data

Enter the `~/IMU_Library` directory and run the `IMU_Serial_Library.py` file

代码块

```
1 cd ~/IMU_ROS2/IMU_Library
2
3 # 运行 IMU 数据打印文件
4 python3 -m IMU_Library.IMU_Serial_Library
```

A terminal window titled 'juxi@juxi-desktop: ~/IMU_ROS2/IMU_Library' showing the execution of a Python script. The script outputs a RuntimeWarning about module loading, followed by system information (Linux), serial port status (opened at /dev/ttyUSB0), and a series of IMU sensor readings including acceleration, gyroscope, magnetometer, attitude angles, quaternions, and barometric pressure.

```
juxi@juxi-desktop:~$ cd ~/IMU_ROS2/IMU_Library
juxi@juxi-desktop:~/IMU_ROS2/IMU_Library$ python3 -m IMU_Library.IMU_Serial_Library
/usr/lib/python3.10/runpy.py:126: RuntimeWarning: 'IMU_Library.IMU_Serial_Library' found in sys.modules after import of package 'IMU_Library', but prior to execution of 'IMU_Library.IMU_Serial_Library'; this may result in unpredictable behaviour
  warn(RuntimeWarning(msg))
当前系统: Linux
IMU串口已打开! 波特率: 115200
成功打开串口: /dev/ttyUSB0
-----数据接收线程已启动-----
固件版本: V1.1.2
数据报告频率已设置为: 10 Hz
加速度计: [-0.04248176519058809, -0.1782280953398236, 0.977568895535142]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-26.85628833887753, 9.375286111026337, -19.77599414044618]
姿态角: [-10.319286290999077, 2.0958281784142407, -167.86505375907623]
四元数: [0.10636502504348755, 0.008655189536511898, 0.09148654341697693, -0.9883670806884766]
气压计: [-0.17, 19.22, 100668.15625, 100666.07812]

加速度计: [-0.04345835749382, -0.17676320688497574, 0.977568895535142]
陀螺仪: [0.0, 0.0, 0.0]
```

Note: The above is the data reading for a 10-axis IMU. The 6-axis has no Magnetometer and Barometer data, and the 9-axis has no Barometer data.

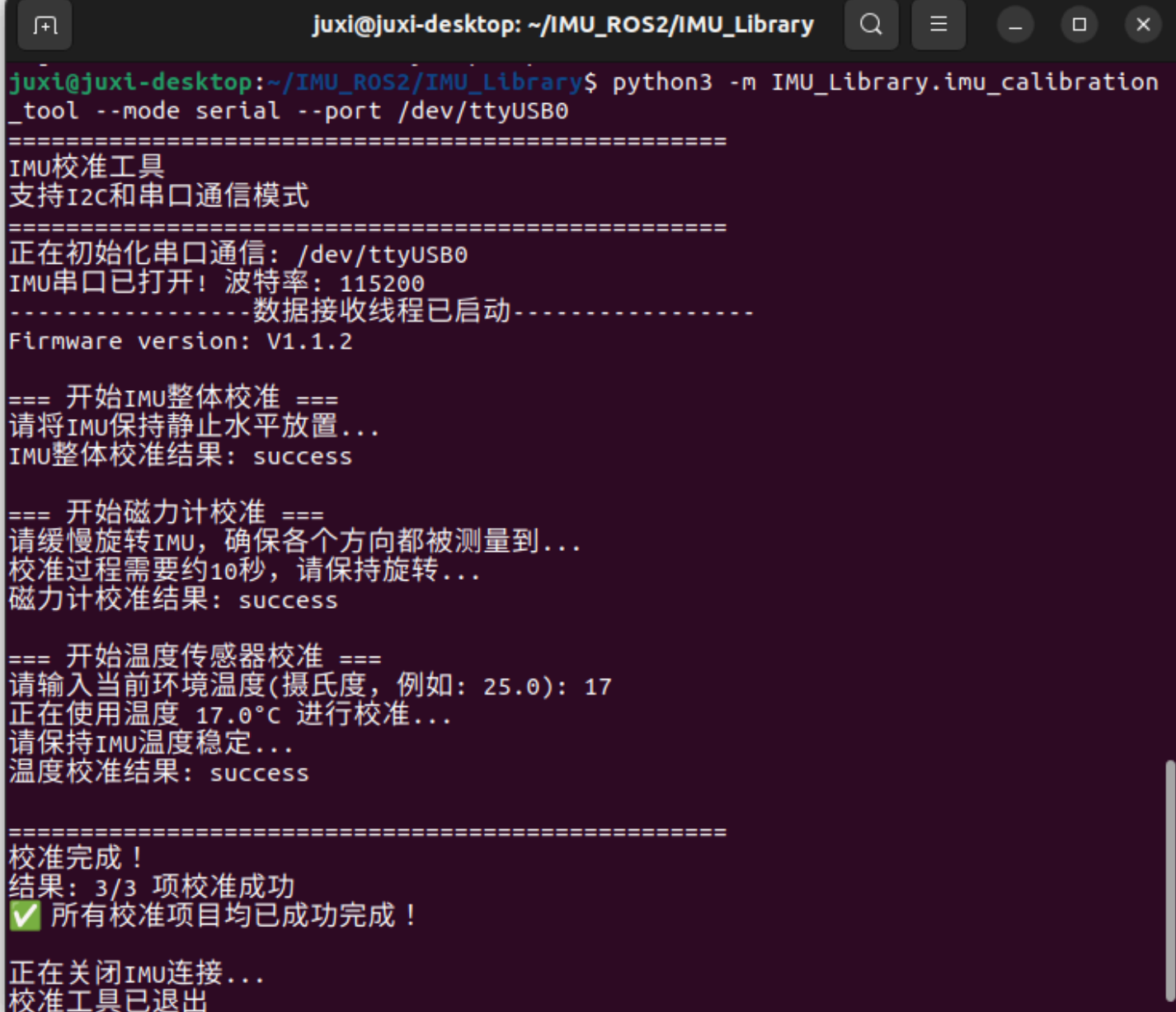
5. IMU Calibration

Enter the ~/IMU_Library directory and run the imu_calibration_tool.py file

代码块

```
1 cd ~/IMU_Library
2
3 # 运行 IMU 校准代码文件 --串口通讯校准
4 # 执行所有校准 (整体、磁力计、温度)
5 python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-serial
6
7 # 仅整体校准
8 python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-serial --calibrate imu
```

```
9
10 # 仅磁力计校准
11 python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-
    serial --calibrate mag
12
13 # 仅温度校准
14 python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-
    serial --calibrate temp
```

A terminal window titled 'juxi@juxi-desktop: ~/IMU_ROS2/IMU_Library' showing the execution of the IMU calibration tool. The user runs the command 'python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/ttyUSB0'. The output shows the tool's initialization, including setting the baud rate to 115200 and starting a data reception thread. It then proceeds with three calibration steps: 1. IMU overall calibration (success), 2. Magnetometer calibration (success), and 3. Temperature sensor calibration (success, using 17.0°C). The process concludes with a summary of successful calibrations and the tool's exit.

```
juxi@juxi-desktop: ~/IMU_ROS2/IMU_Library
juxi@juxi-desktop:~/IMU_ROS2/IMU_Library$ python3 -m IMU_Library.imu_calibration
_tool --mode serial --port /dev/ttyUSB0
=====
IMU校准工具
支持I2C和串口通信模式
=====
正在初始化串口通信: /dev/ttyUSB0
IMU串口已打开! 波特率: 115200
-----数据接收线程已启动-----
Firmware version: V1.1.2

=== 开始IMU整体校准 ===
请将IMU保持静止水平放置...
IMU整体校准结果: success

=== 开始磁力计校准 ===
请缓慢旋转IMU, 确保各个方向都被测量到...
校准过程需要约10秒, 请保持旋转...
磁力计校准结果: success

=== 开始温度传感器校准 ===
请输入当前环境温度(摄氏度, 例如: 25.0): 17
正在使用温度 17.0°C 进行校准...
请保持IMU温度稳定...
温度校准结果: success

=====
校准完成!
结果: 3/3 项校准成功
✅ 所有校准项目均已成功完成!

正在关闭IMU连接...
校准工具已退出
```

6. Precautions

If the device ID can be found but the device number cannot, you can refer to the following command to install the ch34x driver

代码块

```
1 sudo apt remove brltty
```

```
2  git clone https://github.com/clhchan/CH341SER.git
3  cd CH341SER
4  make -j6
5  sudo make install
6  sudo modprobe ch34x
```